MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

1/ TYPE 8827

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Coaxial, tunable frequency 16,410 to 16,625 MHz, pulsed, integral magnet, 50 kw rated peak power output

ABSOLUTE RATINGS:

Parameter:	Ef	if (surge)	If	tk	tpc	rrv	Ib	ib	epy
Unit:	V	a	A	sec	µs	kv/µs	mA	a	kv
Maximum:	4	35	20	---	1.2	180	22	20	16.5
Minimum:	---	---	---	180	0.16	50	---	10	---

ABSOLUTE RATINGS:

				Temperature		Pressurization	
Parameter:	Pi	pi	VSWR	Anode	Input bushing	Input	Du
Unit:	W	kw	---	°C	°C	psia	---
Maximum:	275	275	1.5:1	175	200	25	0.0011
Minimum:	---	---	---	-65	-65	12	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Mounting position: Any
Mounting support: Flange
Cathode: Pressurized (see note 8)
Magnet isolation: See note 12
Cooling: Forced air (see note 20)

Coupling:
Output: Mates with UG-541
Input: Mates with Jettron P/N 90-001-C, or equivalent
Weight: 4.75 pounds maximum (2.154 kg)

TEST CONDITIONS: See note 3

Parameter:	Ef	tpc	rrv	Du	epy	VSWR	T (anode)	T (bushing)	tk
Unit:	V	µs	kv/µs	---	kv	---	°C	°C	sec
Maximum:	2.84	0.55	170	0.001	16.5	1.1:1	165	190	---
Minimum:	2.57	0.45	---	---	15.5	---	-62	-62	180
	(Note 2)	(Note 5)	(Notes 6, 7)	(Note 4)			(Notes 9, 10, 11)		

GENERAL:

Qualification - Required

1/ Replaces Varian type No. BLM-199

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
1006	Salt spray (corrosion)	14	No voltages applied	---	---	---	---
1143	Forced cooling	11,19,20		ΔT	---	75	°C
4027	Temperature coefficient	19	T = 70° to 150°C	ΔF/ΔT	---	0.4	MHz/°C
1042	Shock, specified pulse	15	Condition K; no voltages applied	---	---	---	---
1031	High-frequency vibration	15	No voltages applied	---	---	---	---
1047	High-temperature performance	15,18	No voltages applied	---	---	---	---
1047	Low-temperature performance (1)	15,16	No voltages applied	---	---	---	---
	(2)	15,17	Snap-on	---	---	---	---
4311	Frequency pushing figure	21	ib = 12 to 18 a	ΔF/Δi	---	0.5	MHz/a
	Quality conformance inspection, part 1	22					
1256	Electrode current (anode)	-		ib	5	13	a
4303	Warmup time	-	Ef = 3.5 Vac; tk = 150 sec (min)	If	10	11.5	A
4304	Pulse characteristics	5,6,7		---	---	---	---
4308	RF bandwidth	23		BW	---	1.32/tpc	MHz
4308	Minor lobe ratio	23		Ratio	9	---	dB
4223	Mechanical tuning range	---	Upper limit	F	16,625	---	MHz
			Lower limit	F	---	16,410	MHz
4306	Pulse voltage	---		epy	15.5	16.5	kv
4315	Stability	24		MP	---	0.5	%
4315	Starting stability	24,25		MP	---	0.5	%
4250	Power output	---		Po	50	75	kW
	Quality conformance inspection, part 2						
4310	Frequency pulling figure	---	VSWR = 1.5:1	ΔF	---	15	MHz
	Quality conformance inspection, part 3						
---	Life test	1,5,26	Group D	t	1,500	---	Hours
---	Life-test end points:						
4250	Power output	13		Po	40	---	kW
4308	RF bandwidth	13,21		BW	---	2.5/tpc	MHz
4308	Minor lobe ratio	13,21		Ratio	6	---	dB
4315	Stability	13,17		MP	---	0.5	%

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NOTES:

1. Cycle life tests of 30 minutes "on" preceded by 3 minutes maximum filament schedule and 7 minutes "off" shall be required until the total accumulated "on" time equals the specified life. During the "on" periods, a VSWR of 1.5:1 maximum shall be introduced into external load circuit and the phase adjusted through 22 G during each "on" time.

2. Prior to the application of high voltage, the cathode shall be heated to the required initial operating temperature. This may be done by applying 4.0 volts ±5 percent for 3 minutes. On the application of anode power, the heater voltage should be lowered to the voltage specified. The tube heater shall be protected against arcing by the use of a connector that places a minimum capacitance of 4,000 pF across the heater directly at the input terminals.

3. To relate the various parameters, use the following formula:

$$Pi = ib \times Du \times 16{,}000$$

4. Heater voltage shall be 2.7 V for 0.0007 duty cycle.

5. The characteristic of the applied pulse must be those which result in proper starting and oscillation. The rate of pulse voltage rise, the percentage of pulse voltage ripple, and the rate of pulse voltage fall are among the more important considerations. The cognizant service electron tube group should be consulted regarding pulse characteristics as related to the specific application.

6. The rrv shall be expressed in kv for each μs defined by the steepest tangent to the leading edge of the voltage pulse about 80 percent amplitude. Any capacitance used in viewing system shall not exceed 6.0 pF. No spike or ripple shall exceed ±7 percent of the average peak value of voltage or current. Inverse voltage shall not exceed 20 percent of the forward voltage. The negative amplitude of any post pulse voltage oscillation shall not exceed 10 percent of the average peak value.

7. The rate of rise of voltage (rrv) shall be expressed in kv/μs defined by the steepest tangent to the leading edge of the voltage pulse above 50 percent amplitude. Any capacitance used in viewing shall not exceed 6.0 pF.

8. The mounting plate shall provide for cathode terminal pressurization of 25 ±2 psia and a leak rate not to exceed 0.1 standard cubic inch for each minute.

9. With the specified cooling arrangements and conditions, the anode and input bushing temperatures shall not exceed the specified values.

10. A total airflow of approximately 20 cfm shall be used for cooling unless otherwise specified.

11. The temperature is to be measured at the points indicated on figure 1.

12. In handling and mounting the magnetron, care shall be exercised to prevent demagnetization. A minimum distance of 6 inches shall be maintained between the tube and ferromagnetic materials or energized magnets.

13. If during life test the tube does not meet the specified limits, it shall be recycled for an additional five cycles. At such time the tests shall be repeated. If the tube fails the second test, it shall be considered unsatisfactory.

14. The interior of the waveguide flange, cathode terminal, and portion of the mounting plate adjacent to cathode terminal may be sealed off during this test to simulate the installed condition.

15. After subjection to the specified test, the tube shall meet the requirements for rf bandwidth, minor lobe ratio, and stability as specified.

16. The magnetron shall be stored for 72 hours at a chamber ambient temperature of -62°C (-80°F).

17. The tube shall be allowed to thermally stabilize at -54°C (-65°F), apply standby filament schedule (3 minutes minimum) then turn magnetron on and off several times to assure "snap on" capabilities. Operate for 1 additional hour. The rf envelope shall be monitored to evaluate operation.

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NOTES: -Continued

18. The magnetron shall be stored at +85°C (+185°F). Thermally stabilize for 4 hours.

19. Temperature measurements shall be made only after thermal equilibrium has been reached.

20. A total airflow of 10 cfm at approximately 760 mmHg and 25°C, shall be directed into the fins. The specified rise above ambient temperature shall not be exceeded.

21. The pushing factor shall be measured in three steps, 2 amperes each (10 to 12, 12 to 14, and 14 to 16), and no value shall exceed the limits specified herein.

22. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

23. The magnetron shall be operated into a transmission line with a VSWR of 1.5:1 minimum adjusted in phase to produce the maximum spectrum degradation.

24. Stability shall be measured in terms of the average number of output pulses missing, expressed as a percentage of the number of input pulses applied during the period of observation. The missing pulses (MP), due to any cause, are considered to be missing if the rf energy is less than 70 percent of the normal energy level. The VSWR shall be adjusted to that phase producing maximum instability and the MP counted during the last 3 minutes of a 5-minute test period.

25. Immediately after a nonoperating holding period of 168 hours minimum, the starting stability test shall be performed on 100 percent of the submitted lot.

26. Life test shall be conducted in accordance with the following cycle:

Condition	Ib	Ef	Duration
Standby	0	3.5	3 minutes
Test 1	13.5 ma	2.7	30 minutes
Off	0	0	7 minutes

27. Revision letters are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N055)

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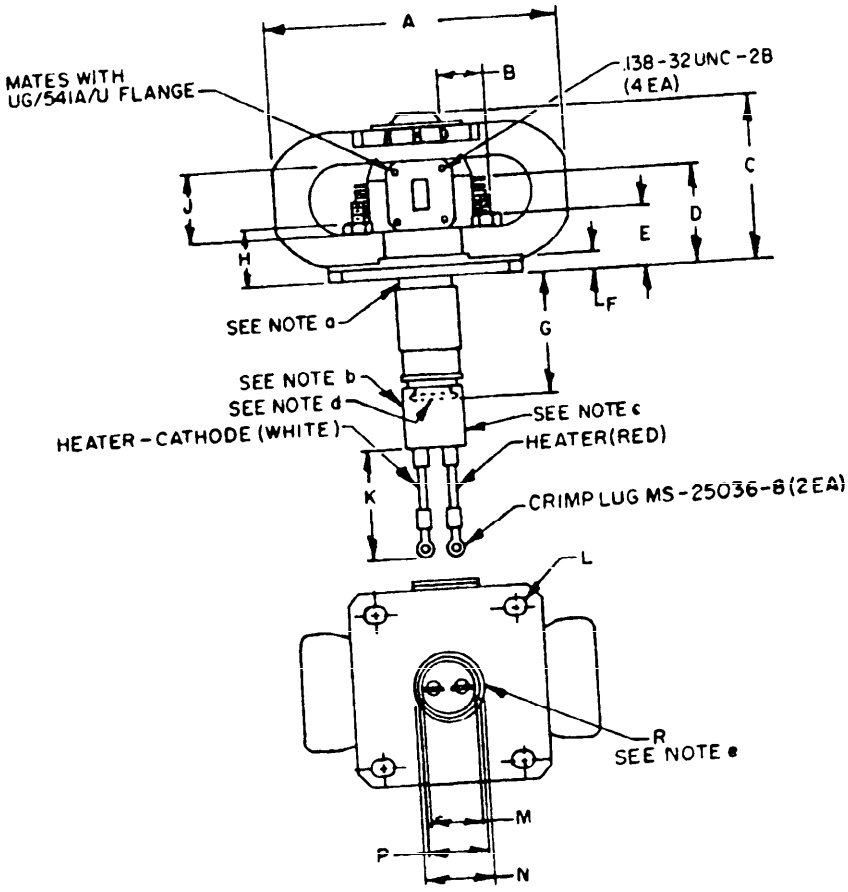


FIGURE 1. Outline drawing of electron tube type 8827.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Qualification inspection			
A	---	5.800 (147.32)	
C	---	3.000	
G	---	2.190	
M	---	1.100	
Quality conformance inspection, part 1			22
B	0.866	0.926	
D	---	1.922	
E	0.995	---	
F	---	0.250	
L	0.129 R	0.141 R	
Quality conformance inspection, part 2			
H	0.980	1.020	
N	1.745	1.755	
P	1.345	1.355	
R	0.111	0.121	
Nominal			
J	1.312		
K	2.312		

NOTES:
a. Anode temperature shall be measured at this point.
b. Bushing temperature shall be measured at this point.
c. Jettron Products, Inc. connector No. 90-001, or equivalent.
d. Protective enclosure.
e. This dimension applies to depth of "O" ring groove.

FIGURE 1. Outline drawing of electron tube type 8827 - Continued.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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